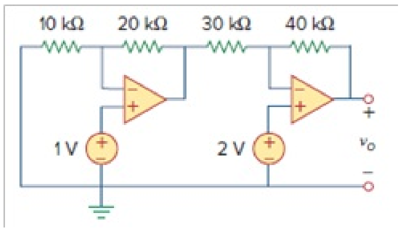
Chapter 5, Problem 78P

Problem

Use *PSpice* or *MultiSim* to obtain *vo* in the circuit of Fig. 5.101.

Figure. 5.101:



Step-by-step solution

Step 1 of 4

Refer to Figure 5.106 in the text book.

Consider the bridge circuit shown in Figure 1.

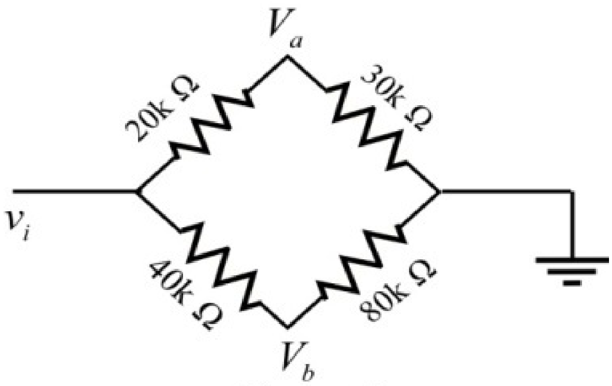


Figure 1

Step 2 of 4

Express the node volatges v_a and v_b in terms of the input volatge v_i.

$$v_a = \left(\frac{30k}{30k+20k}\right)v_i$$
$$= \frac{3}{5}v_i$$

$$v_b = \left(\frac{80k}{80k+40k}\right)v_i$$
$$= \frac{2}{3}v_i$$

Indicate the node volatges in the circuit and draw the equivalent circuit.

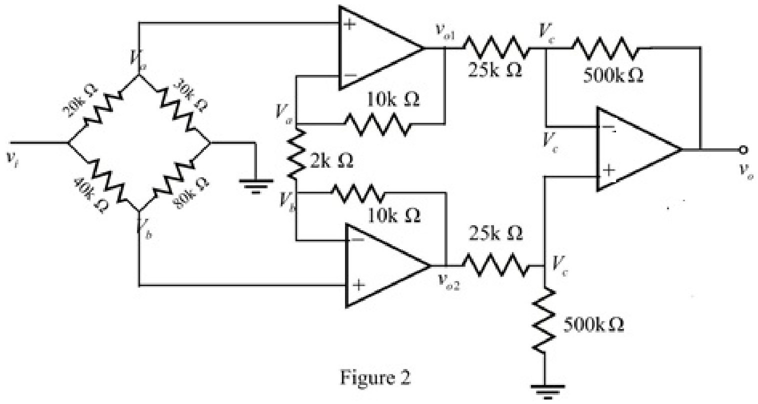


Figure 2

Step 3 of 4

Apply Kirchhoff's voltage at node voltage v_a.

$$\frac{v_a - v_{o1}}{10k} + \frac{v_a - v_b}{2k} = 0$$
$$6v_a - v_{o1} - 5v_b = 0$$
$$6\left(\frac{3}{5}\right)v_i - v_{o1} - 5\left(\frac{2}{3}\right)v_i = 0$$
$$v_{o1} = \frac{4}{15}v_i$$

Apply Kirchhoff's voltage at node voltage v_b.

$$\frac{v_b - v_a}{2k} + \frac{v_b - v_{o2}}{10k} = 0$$
$$-5v_a + 6v_b - v_{o2} = 0$$
$$-5\left(\frac{3}{5}\right)v_i + 6\left(\frac{2}{3}\right)v_i = v_{o2}$$
$$v_{o2} = v_i$$

Step 4 of 4

Use voltage division principle to find the expression of v_C.

$$v_C = \left(\frac{500k}{500k + 25k}\right) v_{o2}$$
$$= \left(\frac{500}{525}\right) v_i$$

Apply Kirchhoff's current law at inverting terminal of output op amp.

$$\frac{v_c - v_{o1}}{25k} + \frac{v_c - v_o}{500k} = 0$$
$$21v_c - 20v_{o1} = v_o$$
$$21\left(\frac{500}{525}\right) v_i - 20\left(\frac{4}{15}\right) v_i = v_o$$
$$14.667 v_i = v_o$$

$$\frac{v_o}{v_i} = 14.667$$

Thus, the gain $\frac{v_o}{v_i}$ is $\boxed{14.667}$.